

Position Title and No.	Team Leader
Name of Expert:	Rajinder Khazanchi
Date of Birth:	1959
Country of Citizenship/Residence	Indian

Education:

Names of educational institutions	Dates attended	Degree(s)/Diploma(s) obtained
Indian Institute of Technology, Delhi, India	1987	M. Tech Degree in Structural Engineering
Indian Institute of Technology, Delhi, India,	1981	B. Tech Degree in Civil Engineering

Employment record relevant to the assignment:

Period	Employing organization and your title/position. Contact info for references	Country	Summary of activities performed relevant to the Assignment
June 2013 to Present	Employer: Synergetics Management & Engineering Consultants Positions Held: Director Contact Reference: Sandip Kaul – Director +919811087507	India	Responsible for running a small advisory firm covering areas of Ports and maritime, IWT, Sea logistics and berthing facility design. Key projects – Hazira port valuation for AMNSI, Plant – market Logistics for Cement majors, Project Director for - A 10000 T ship lift facility on East coast for Indian Navy and as Team leader for a World Bank funded IWT passenger terminals project in Bangladesh. Amongst the main work assignments – have provided technical coverage on a dry port project in Egypt, a cruise terminal project in Cambodia, two container terminals for a private developer on East and West coast of India.
Jan 2017 to Nov 2017	Employer : Howe Engineering Projects India Pvt Ltd. (part of Adani group) Positions Held : Chief Operating Officer Contact Reference: Dr.Malay Mahadevia, Group President - +919825016638		Responsible for the project coordination and delivery of the portfolio of engineering and project management assignments dealing with ports, inland waterways (NW-1) and railways. And the business development activities.
Feb 2012 to May 2013	Employer: AECOM India Pvt. Ltd. Positions Held: Executive Director Contact Reference: Avinash Misra – CEO Mo. +919910855998	India	Head of Urban Infrastructure development group within the Water and Urban Development business line for the Indian sub-continent. Responsible for acquisition of business, team development, project deliveries, cash collections for the group. Besides regular business turnover of 1.5 Million USD, have been key member of the large program management win for New Cities project - Dholera Special

			Investment Region in Gujarat covering services over 5 years with a fee of 28 Million USD for an initial phase of 2 Billion USD development focused over 500 sq.km for a semi - government entity Delhi Mumbai Industrial Corridor development corporation (DMICDC). Also provided Senior management coverage for the Ports and Maritime division for a major program management bid for a Naval Base on the West coast of India
Apr 2009 to Jan 2012	Employer: Synergetics Management & Engineering Consultants & Synergetics Logistics Solutions Pvt. Ltd. Positions Held: Director Contact Reference: Sandip Kaul – Director +919811087507	India	Was responsible for covering high end services like business advisory and strategic planning. Provided Technical and project management advice for the development of a 10 MTPA Iron Ore terminal at Paradip Port for consortium of the Noble Group (Hong Kong) and Gammon Infrastructure Ltd. Client was successful in its bid for the development of the terminal at estimated cost of US \$ 110 Million. Delivered 2 more consulting assignments – one dealing with the due diligence for a Private Equity Fund for the development of a Green field port in Orissa state and second one dealing with the site selection study for a coast based thermal power plant for a large industrial group covering the state of Gujarat. Besides these assignments developed a multi-modal transport chain for 2 cement companies covering end to end movement from their factories in Gujarat to markets in around Mumbai.
March 2008 to February 2009	Employer: Mundra Port and SEZ Ltd Positions Held: President Contact Reference: Dr.Malay Mahadevia, Group President - +919825016638	India	Headed the SEZ business vertical for the development of a 100 sq.km special economic zone at the Mundra Port in Kutchh region of Gujarat state. Responsible for infrastructure development works of over 150 Million USD covering infrastructure projects to support the development of India's largest Special economic zone. Was also on board of a couple of subsidiary companies linked with the SEZ – a 125 acre Textile park (MITAP), and the local utility distribution companies covering supply of electricity and water and waste water. Prepared business plans for these utility companies including development of standard operating procedures, tariff and commercial conditions. This job has provided good experience

			in managing a diverse team of over 80 persons comprising of marketing, engineering services, regulatory & legal, corporate liaison, construction management, estate, security and finance.
June 2005 to February 2008	Employer: Haskoning India Positions Held: Chief Executive Officer Contact Reference: Simon Harries – Division Director +447747638817	India	During my tenure, the Indian operations were grown from about 5 persons to over 80 + in a period of 2.5 years. With most of the design and project management work being carried out internally. Consulting operations which were initially focused on Maritime Sector, were expanded into other areas like Architecture and Buildings, Industrial Installations and specialized structural engineering linked to design of LNG tanks. Growth plans were put in place for adding capability in all the other divisions of the parent company - Environment, Water resources and Infrastructure. The Indian operations were successfully integrated with the groups divisional structure as prevalent in Europe
January 2000 to May 2005	Employer: Synergetics Management & Engineering Consultants Positions Held: Director Contact Reference: Sandip Kaul – Director +919811087507	India	Was a director of this consulting team and worked on several projects in the role of an Infrastructure specialist having design, financing and project management experience in the areas of Ports, Road, Rail, Ropeways and Water transport. Coordinated work for thermal power project ----- an EPC bid for 2 x 30 MW Power plant and a Feasibility study an LNG terminal on East Coast of India
June 1997 to December 2000	Employer: HASKONING India Pvt. Ltd Positions Held: Director Contact Reference: Simon Harries – Division Director +447747638817	India	Set up the subsidiary and grew the team along with acquisition of a couple of major design engineering and project management jobs in the Maritime and Water Supply sector. Major success was a Project Management job for a Naval base with a fee of over 5 Million Euro. Successful in forging partnerships with local consulting firms, establishing brand awareness and managing client relations.
April 1989 to March 1991	Employer: Pomagalski S.A.(Cable Transport Systems), Grenoble, France Positions Held: Project Manager Contact Reference: Michel Chaudan, Project Director - +33479069363	France	This was a period of working with an International company learning interpersonal skills of working with teams coming in from diverse cultures. Learnt the skills of designing and integrating dynamic systems – a bit about electro-mechanical aspects, testing & commissioning of plants

April 1987 to March 1989	Employer: RITES, Delhi, Transportation and Management Consultancy Positions Held: Manager Contact Reference: Bimlesh Kumar, Group General Manager - +919818311542	India	During the 2 years with one of the largest consulting company in the Indian market operating in the semi-government format, it was a period of honing contract management skills and learning about various other transportation fields including railway and ropeway systems
January 1981 to March 1987	Employer: Consulting Engineering Services (I) Pvt. Ltd., Delhi Positions Held: Senior Engineer	India	Moved through the role of a design engineer to a project manager acquiring skills of working with large multi-disciplinary teams. Besides honing the skills of a structural designer, I learnt about transport economics, project financing, market research and forecasting techniques. Hands on working experience gained during this period has turned out to be mainstay of my working philosophy.

Membership in Professional Associations and Publications:

- Fellow of Institution of Structural Engineers, India
- Member of the Institution of Engineers, India
- Member of the Consulting Engineers Association of India

Training & Publications

- Paper illustrating the experiences and challenges of developing India's largest Special Economic Zone (130 sq.km) at Mundra, Kutchh district, Gujarat presented at Urban Development Conference in Delhi in Aug. 2008.
- Paper on Marketing strategy for the Mundra SEZ (130 sq.km) presented at the conference on Infrastructure development under the aegis of Ahmedabad Management Association – Apr 2008
- Paper on Growth prospects of Container transport in India in the International Symposium on Ports and Infrastructure in Colombo – 2007.
- Paper on Cable propelled Transport systems for Urban Transport – with case studies for Delhi and Kolkata – presented in International Conference on Ropeways in Delhi – 1989
- Visiting faculty member at the Management Development Institute – Gurgaon – lecturing on Project Management courses for senior management from various public and private enterprises

Language Skills (indicate only languages in which you can work):

	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
Hindi	Excellent	Excellent	Excellent
French	Fair	Fair	Fair

Countries of Work Experience:

- India, Bangladesh, Cambodia, Tanzania, Mauritius, France, Netherlands, United Kingdom

Adequacy for the Assignment:

Detailed Tasks Assigned:	Work Undertaken that Best Illustrates Capability to Handle the Tasks Assigned
	Project Name : Plant to Market sea transport of Cement. Year : Oct 2021- Mar 2022 Client : Saurashtra Cement Ltd. Location : West Coast of India Position Held : Team Leader Main features : Logistics study for movement of 0.5 Million T of cement from Plant near Porbandar port to Mumbai region covering evaluation of existing transport chain and design of alternative options using bulk cement shipment including layout of unloading terminal at market end. Activities performed: Assimilation of existing costs of transport chain – plant to port using bagged and bulk cement in FIBC, loading & unloading and storage, ship loading sea transport, unloading at Mumbai port and other 3 alternate sites. Identification of return cargo – fly ash / slag, shipping logistics options using self discharging cement carriers / alternative options using continuous screw / pneumatic unloading, chartering versus acquisition, preliminary layout and design of terminal facilities and costing – capital + operating costs and its impact on unit transport costs. Project Name : Valuation of Port Assets . Year : Jul 2021- Dec 2021 Client : MTBS (Netherlands) . Location : Hazira river terminal, West Coast of India Position Held : Port Specialist Main features : Valuation of the existing port facilities operated by Essar Bulk terminals at Hazira for the buyers (AMNSI) who have acquired the adjacent steel plant of Essar Steel. Existing port assets in terms of the quay length of over 700 m for deep water facilities handling Post Panamax and Cape size vessels upto 15 m draft with associated capital and mtc dredging, reclaimed back up storage and material handling plant of over 25 MTPA. Activities performed: Asset registers, project completion drawings, Depreciation schedules, Ship arrivals and queuing, Port handling performance data review, capacity assessment, BOOT agreement reviews, GMB policy review, Dredging stages and costs spent, diligence of cost numbers based on Development cost data of similar projects. Adaptation of future traffic growth and capital investment and opex. Alternative techniques of valuation using Present value of assets and discounted future cashflows. Inputs on technical aspects including capital and operating costs, condition assessment of port assets, master plan for future capacity expansion and its costs. Alternative port facilities assessments

	Project Name : Ship lift of 10000 T capacity. Year : May 2020- Sept 2021 Client : Directorate General of Naval Projects , India Location : East Coast India Position Held : Project Director Main features : Feasibility study and Detailed Engineering and tender documents for a 10000 T lift capacity ship lift for the Indian Navy on East coast. Head of the team of Principal consultants Trafalgar EPC Ltd - India along with Ship lift specialist Inros Lackner, Germany. Activities performed: Coordination of the multi-disciplinary team including interactions with the DGNP's project team, evaluation of design criteria based on client requirements, alternative layouts and construction techniques, considerations of longitudinal and transverse transfer arrangements, Very high covered workshop bays, work flow planning and sizing of manpower requirements, utilities requirement and design, costing, client presentations, acceptability of options, and preparation of tender documents Project Name : River terminals for Passenger transport, Bangladesh Year : Dec 2019- Sep 2020 Client : Bangladesh Inland Waterway Transport Authority Location : Bangladesh Position Held : Team Leader Main features : Feasibility study, Detailed Engineering and tender documents for 4 passenger terminals along waterways in Bangladesh at Shasanghat (Dhaka), Narayanganj, Chandpur and Barisal and a training institute – DEPTC at Narayanganj under World Bank funding. - Annual Passenger handling capacity of over 90 Million. Activities performed: Planning and design of terminal facilities based on traffic forecasts Feasibility, Detailed designs, cost estimates, tender documents . Project Name : Dry Port (Inland Container terminal) at 6th October city, Egypt Year : Apr 2019 – Dec 2019 Client : Consortium of Container Corporation of India, Port of Singapore Authority and HA – Egypt as associate of MTBS- Netherland Location : Egypt Position Held : Terminal designer Main features : Technical bid for the development of the railway linked Dry port in Egypt with initial capacity of handling 120000 TEU /annum growing up to 800,000 TEU per annum over a project site of 42 HA. Activities performed: Planning, layouts, and design of the Dry port terminal with
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	railway linkage to main ports of Egypt, design of Customer facilitation centre, customs offices, Container stackyard, Railway siding and unloading – loading arrangements, designs of Container freight station, stuffing – de-stuffing warehouse, entry gates, security systems, IT integration, capital costs and operating structure. Project Name : Cruise Terminal at Port Sihanoukville , Cambodia Year : Dec 2018 – Mar 2019 Client : Sihanoukville Autonomous Port Authority Location : Cambodia Position Held : Port Expert Main features : Feasibility study for conversion of an old jetty into a terminal for Cruise ships and a sea front development as an add on to improve viability of the project. Activities performed: Planning and design of terminal facilities based on traffic forecasts and identifying the prospective development agencies. Seafront development was added to improve the project viability and private participation. Project Name : Development of 5 Terminals on River Ganga – Varanasi to Haldia, India Year : Jan 2017 – Dec 2017 Client : Inland Waterway Authority of India Location : River Ganga - India Position Held : Project Director Main features : Detailed Engineering phase of the project involving design and tender documents for 2 multi-modal terminals and 10 RO-RO cum passenger facility terminals Activities performed: Guidance for the team and client presentations and approvals for the Planning and design of terminal facilities based on traffic forecasts and the projected ferry vessels proposed for use on the river system. Main challenges being large water level variations – both horizontal and vertical, incorporating vehicle segregation and passenger flows and parking areas, ticketing and security solution for the terminals, selection of green field locations, bank stabilization and implementation of other river training and channel improvement measures including capital and maintenance dredging solutions. Project Name : Multi-purpose cargo terminal at Paradip Port, India Year : Mar 2016- Jan 2017 Client : International Cargo Terminals And Rail Infrastructure Pvt. Ltd Location : India Position Held : Technical Advisor
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	Main features : A Jetty of 450m long with dredged depth of 17 m for handling vessels up to 15 m draft to handle clean cargo comprising of Containers, Steel coils and fertilizer with adjoining container yard of 48000 sq.m and hard stand for steel coils stacking of 12200 sq.m and attached fertilizer storage warehouse of 21000 sq.m with mechanized loading, bagging and train dispatch facilities including railway siding for 5 rake rakes of 650m including Gate system. Being developed on Public Private Partnership basis. Activities performed: Review of the Planning and design of terminal facilities, designs and tender process for award of work for the Main berth and the back up facilities and yard. With value engineering moved the tender process from the consultants designs to Design build option for the Main berth leading to a cost reduction of over 10 Million USD. Project Name : Container terminal at Kandla Port, India Year : Mar 2016-Jan 2017 Client : International Cargo Terminals And Rail Infrastructure Pvt. Ltd Location : India Position Held : Technical Advisor Main features : On an existing brown field development , old berth of 545 m was refurbished with repair of the Gantry crane tracks, cable trenches and retro-fitting of old quay cranes and a new container yard of 150,000 sq.m was added along with the Customer facilities building, Gate house and Railway siding area 43000sq.m paved surface. Project developed on Public Private Partnership basis. Activities performed: Planning of Geotech investigations, design reviews for the pavement yard and technical guidance to the Project management team and the engineering team. Work planning and progress reviews. Project Name : Port Kasanga on Lake Tanganyika Year : 2011-12 Client : Tanzania Port Authority Location : Tanzania Position Held : Project Director Main features : A Jetty on lake for handling up to 2 cargo barges and passenger ferries with associated handling facilities including passenger amenities, storage ware houses, open storage areas, gates, security system and IT integration. Project to be developed on Public Private Partnership basis. Activities performed: Planning and design of terminal facilities based on traffic forecasts and identifying the prospective development agencies including stake holder communications and preparation of the PPP bidding documents. Project Name : <i>4 P Logistics company</i> Year : 2009 -11 Client : Saurashtra Cements Ltd. And Gujarat Sidhee Cements Ltd Location : Porbandar – Mumbai, India Position Held : Business Head / Promotor
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	Main features : Factory to Market logistics chain for cement companies from plants in Saurashtra region of Gujarat to Markets in Mumbai. with containerization and coastal shipping through Porbandar Port. Chain set up to move over 50,000 TEU through the container terminal and yard established at Porbandar Port Activities performed: Evolving a business plan, fund raising, marketing, start up including setting up supply chain contracts, identifying sub vendors, setting up container terminal at Porbandar port, hiring in trailers on both ends of chain, establishing IT system - for information flow and tracking, chartering container ships, leasing containers, Billing systems, etc. Managed to get contracts worth over 100 cr. Per annum and started the first weekly container service to Mumbai ICT. Managed last mile connectivity and integrated operations with client MIS systems. Project Name: 10 MTPA Iron ore terminal Year: 2009-10 Client: Blue Water Iron Ore Terminal Pvt. Ltd. Port Trust Location : Paradip port, East Coast, India Main project features: Iron Ore terminal with Berth length of over 350 m and associated loading equipment of travelling loaders of 6000 TPH a and back end connectivity through cross country conveyors of matching capacity, stacker reclaimers in stock pile area and rail unloading system comprising of 2 nos unloaders of 1500 TPH capacity. Positions held: Project Director - Port Planner Activities performed: for the advisory services linked to PPP bid by consortia of Noble Group – Hongkong, Gammon Infra and MMTC – India Assignment included review of the layouts, cost estimates, time frame and operating costs of the berthing and handling facilities. After the consortia winning the bid, acted as the project director for the early start-up activities including identification of design engineers, site management and project management team. Name of assignment or project: Site selection study for a 1320 MW Power plant Year: 2009 Location: India Client: Jindal Group Positions held: Project Director Main project features: Site selection along the Gujarat coast Activities performed: Sizing of port side facilities and material handling systems and assessment of capital and operating costs with a view to establish charge factor for port costs. These would form part of the coal transport costs for the imported coal including the shipment and the entire chain costs to establish least cost option for the site selection besides compliance with other criteria for site selection. Project Name: Preparation of Business Plan for Kolkata Port Year: 2006-07 Client: Kolkata Port Trust Location: India Main project features: Future development and growth of Haldia – Kolkata port system envisages a concept of riverine port where by it has been proposed to integrate Inland water transport as a major initiative along with offshore trans-
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	shipment terminals. Positions held: Team leader - Port Planner Activities performed: Activities included review of existing master plans, preparation of future development plans based on traffic projections and preparation of cost estimates for inputs for financial analysis Name of assignment or project: Due Diligence of a Shipyard in Goa Year: 2007 Location: India Client: Private sector client Positions held: Lead Shipyard Specialist Activities performed: Valuation of existing infrastructure, cost estimates and financial analysis including review of balance sheets Name of assignment or project: Rehabilitation of an existing jetty of GMB Year: 2005-06 Location: India Client: Welspun Industries Main project features: Preparation of detailed specifications and detailed engineering Positions held: Project Director Activities performed: Review of stability of existing facility, strengthening and rehabilitation of the jetty Name of Services or Project: Development of Cruise Terminal at Port Louis Year : 2006-07 Location : Mauritius Client: Port Authority of Mauritius Main Project Features: Master planning, preparation of lay-outs Position held : Port Planner Activities Performed Assessment of requirement of surveys and investigations, detailed design and engineering. Name of Services or Project: Development of Harbor at Two Islands Year : 2006-08 Location : Andaman and Nicobar Client: Public works department, Andaman and Nicobar Islands Main Project Features: Master planning, conceptual and detailed design Position held : Project Director/Senior Structural Engineer Activities Performed: Alternative layouts, design criteria, land and marine surveys, evaluation of data and detailed designs, cost estimates, preparation of tender documents Name of assignment or project: Planning and design for a proposed 2nd container terminal at Chennai Port Year: 2005 Location: India Client: Private developer Positions held: Project Leader Activities performed: BOT bid preparation Name of assignment or project: Dahej Port Year: 2006 Location: India Client: Adani Positions held: Project Leader Activities performed: Development of Material Handling System and Port Back-up areas
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	Name of Project: Breakwater design for Gangavaram Port Year : 2004 Location : India Main Project Features: Bid for design-build contract Client: Leading Maritime contractor _ HCC Ltd. Position held : Port Design consultant Activities Performed: Planning and design of two breakwaters. Name of Project: Development of CFS at Ennore Year : 2004 Client: SICAL-CWT Position held : Project Manager Activities Performed: Responsible for assessment of infrastructure required to manage / handle container traffic for a 10-year time horizon; assessment of the requirement of container handling equipment; preparation of layout plan, ball-park cost estimates; etc. Name of Project: Due diligence of the container terminal at Pipavav Port Year : 2003-04 Location : India Client: IDFC Asset Management Company Main Project Features: As part of expansion program of the port – deepening of the harbor basin and approach channel from 11 m depths to 14 m was envisaged along with the construction of 700m long container berth and back up terminal facilities. The asset management company wanted a diligence report as part of its decision to invest in the equity of the port. Position held : Project Leader Activities Performed: Review of the planning criteria and dredging data to confirm the facility requirements and costs thereof. Also a review of project risks and time and costs update. Name of Project: Development of a Private Port in Gujarat Year : 2003-2004 Location: India Client: Dholera Port Pvt. Ltd. Main Project Features: Development of a green-field port at Dholera in the Gulf of Khambhat. The project envisaged development in 2 phases with initial phase development for 17 Million Tonne per annum (MTPA) capacity and there after future extension into a 35 MTPA. Main purpose was the divestment of the port development rights. Position held : Project Director Activities Performed: Responsibilities include vetting of the Detailed Project Report and other planning and environmental documents, reviewing the financial structuring of the project, identification of strategic partners, inviting proposals from them for the participation in the development of project and finalizing the partnership agreement. Name of Project: The design of a multi-purpose bulk berth for loading Cement and Clinker Year : 2003-2004 Location: India Client: Gujarat Sidhee Cement Ltd Main Project Features: Located at Vadodara Jhalla close to Veraval on the Suarashtra coast line in Junagarh District of Gujarat,, this project envisaged 2 multi-purpose berths to handle coal and clinker with additional facilities for export of bulk cement and bagged cargo. Traffic potential of 2.5 MTPA in ph. 1 increasing to 4 MTPA.
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	Position held : Project Manager Activities Performed: As Project Manager responsible for the detailed Engineering, Tendering of a 300m long multi-purpose berth with 300 m long approach jetty Name of Project: Adani Mundra Port- Earthquake damage review and rehabilitation Year : 2001-2002 Location : India Client: Royal Sun Alliance Main Project Features: Earthquake damage review and rehabilitation Position held : Port structural expert Activities Performed: reviewed existing design, inspection of damage records and assessment of cause –damage correlation followed by review of repair / rehabilitation scheme for the affected marine works and shore side tankage. Work area covered of an approach trestle —550 m x 12 and two main jetties 31 m x 216 m and 31m x 165 m plus a connecting jetty of 202m x 7m and an approach bund of 700 m length for the marine works followed by tank farm area with tankage in excess of 265,000 KL. Name of Project: Cement bulk terminal of 1 Million TPA Year : 2001 Location : India Client: Saurashtra Cement Ltd Main Project Features: Bringing bulk cement for the cement storage terminal Position held : Port planner Activities Performed: As Port Planner responsible for the layout, design and costing of berthing and handling facilities for ships of 3 – 5000 DWT size bringing bulk cement for the cement storage terminal. Estimated cost of the terminal – Rs. 380 million. Name of Project: Dry Port Project in Rajasthan – for PDCOR /RIICO under funding from International Development Agency Year : 2000 Location : India Client: West Bengal Pollution Control Board, Kolkata Main Project Features: Feasibility study for the setting up of Dry Port in the state of Rajasthan. Position held : Design Engineer cum terminal Planner Activities Performed: Responsible for the master-planning and design of various facilities at the proposed Container terminal including preparation of cost estimates. Name of Project: Project Coordination for 2 x 30 MW Power project Location: India Year: 2000 Client: Utkal Alumina Main Project Features: Coal based steam and power plant Position held : Dy. Project Manager Activities Performed: Responsible for the preparation of an EPC bid in coordination with the Belgian parent and domestic sub-contractors and vendors. Name of Project: Naval Base at Karwar, Karnataka Year : 1999 Location : India Client: Indian Navy Main Project Features: the Construction Management of a Naval Base at Karwar, Karnataka for the Indian Navy with a tendered price of Rs. 6000 million Position held : Deputy Project Manager and Port Planner Activities Performed: Responsible for the planning, progress review and contractual
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	aspects of marine contract comprising of 4.1 cu.m of quarry works for the breakwaters and land reclamation works together with 15 million cu.m of dredging Name of Project: Port Craft Jetty at Kandla Port, Year : 1998- 1999 Location : India Client: Kandla Port Trust Main Project Features: Construction of Jetty Position held : Project Director Activities Performed: Project Director for the planning, design, engineering and construction supervision for a Jetty in tidal range of over 7 m. to allow berthing for various port craft – pilot launches, Tug boats, barges Name of Project: Logistic Study for setting up Pig Iron and Coke Oven project on the Gujarat Coast Year : 1996- 1997 Location : India Client: Gujarat Fluoro-Chemicals Main Project Features: Logistic Study for setting up of a Pig Iron and Coke Oven project on the Gujarat Coast for Gujarat Fluoro-Chemicals Position held : Project Director-cum-Planner Activities Performed: Involved in multivariable optimization exercise involving logistics study for raw materials (sourcing) for iron ore fines, lumps and coking coal. Alternative modes of bulk transport like shipping, rail and road transport. Name of Project: Feasibility Study for Augmentation of Ship Repair Facilities Year : 1995-1996 Location : India Client: Uran Ship yard Main Project Features: Layout and design of ship lift system of 4000 T lift capacity as part of the expansion program of Uran Shipyard Ltd. Position held : Design Engineer Activities Performed: Responsible for the layout options and structural design of Ship lift and repair berths in association with naval architectural firm SEA SYST Name of Project: Detailed Engineering and Construction Management of a Jetty and Harbour works Year : 1995-1996 Location : Vadodhara Jhalla on the Saurashtra Coast, India Client: Gujarat Sidhee Cements Ltd. Main Project Features: Detailed Engineering and Construction Management of a Jetty and Harbour works including material handling of bulk cement, clinker and coal for ships upto 60,000 DWT for the Gujarat Sidhee Cements Ltd. Position held : Project Manager-cum Design Engineer Activities Performed: Design of Breakwater and berthing facilities, cost estimates, prequalification of contractors, tender document preparation, floating and evaluation of bids for client. Name of Project: Feasibility Study for Augmentation of Ship Repair Facilities at Gogha Shipyard, Year : 1994 Location : India Client: Alang Marine Ltd Main Project Features: Design of ship lift system of 6000 T lift capacity as part of the expansion program of Alang Marine Ltd.. Position held : Design Engineer Activities Performed: Responsible for the layout options and structural design of
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	Ship lift and repair berths including reclaimed foreshore area including workshops and other infrastructure in association with naval architectural firm SEA SYST Name of Project: Feasibility Study for the setting up of Multi-purpose Berth Year : 1994-95 Location : India Client: Saurashtra Cements Ltd Main Project Features: Feasibility Study for the setting up of Multi-purpose Berth at Porbandar Port Position held : Project Manager-cum-Port Planner Activities Performed: handling vessels up to 40,000 DWT for cement, clinker and coal including material handling and storage system. Name of Project : Feasibility Study for Bulk Cement Transport from Porbandar & Veraval Ports in Gujarat Year : 1993 Location : India Client: Saurashtra Cements Ltd. Main Project Features: Feasibility Study for Bulk Cement Transport from Porbandar & Veraval Ports in Gujarat to Coastal destinations of Bharauch, Mumbai, Goa and Cochin for Saurashtra Cements Ltd. Position held : Project Manager-cum-Port Planner Activities Performed: Feasibility Study Name of Project: Logistic Study for design of Port Facilities and Shipping System for Slag Cement Grinding Units at Haldia & Cochin Port Year : 1993-1994 Location : India Client: Saurashtra Cements Ltd. Main Project Features: Logistic Study for design of Port Facilities and Shipping System for Slag Cement Grinding Units at Haldia & Cochin Port for Saurashtra Cement. Position held : Design Engineer Activities Performed: Logistic Study for design of Port Facilities and Shipping System Name of Project: Design and Site Selection for the Ship Inspection and Repair Facilities Lakshadweep Islands Year : 1992-93 Location : India Client: F.C. Deweger International Consultants Main Project Features: The Ship Inspection and Repair Facilities (ship lift and slip way) for the Inter-Island High Speed Ferries for the Lakshadweep Islands Position held : Team Member-cum-Design Engineer Activities Performed: Design and Site Selection Project Name: Feasibility Study for transport linkage to Paradip Port and Satellite Port at Dhamra Year: 1987-1989 Location: India Client: Central Electricity Authority, Main project features: Feasibility Study for transport of 50 million tonnes of coal along the Brahmani River System by IWT as part of transport linkage to Paradip Port and Satellite Port at Dhamra for further transportation by colliers to the proposed thermal power stations located along the coastline of India. Positions held: Team Member-cum-Planner Activities performed: As Team Member-cum-Planner for Feasibility studies, Designs and Contract Administration of several large projects in the areas of Ropeways and cable transport systems in Auli, Joshimath, Shimla, Himachal Pradesh & J&K state.
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	Project Name: Feasibility study for the transport of 15 million tonnes of coal along the Brahmani-Mahanadi River System in Orissa State Year: 1986-1987 Location: India Client: Central Electricity Authority, Main project features: Feasibility study for the transport Positions held: Team leader - Port Planner Activities performed: As Project Manager-cum-Planner for a Feasibility study for the transport of 15 million tonnes of coal along the Brahmani-Mahanadi River System in Orissa State, India, by Inland Water Transport up to Paradip/Dhamra Port as part of linkage to North Madras Thermal Power Station (NMPTS) for the Tamilnadu Electricity Board. Project Name: System Study for the Planning Commission to evaluate the prospects of development of IWT in the country Year: 1985-1986 Client: Planning Commission of India – Transport Division Location: India Main project features: A country wide initiative covering assessment of navigability and traffic potential for development of inland waterways leading to declaration of on 10 major national waterway systems in India Positions held: Team leader - Port Planner Activities performed: As Team Leader-cum-Planner, carried out studies related to evaluation of navigability of all the waterways in the country, the fleet selection for these, development of modern terminal layouts, interface transfer facilities and other system developments and the cost estimates for these along with cash flows and IRR's etc to prioritize the development of these waterways Project Name: Development of Terminals for the Central Inland Water Corporation (CIWTC) Year: 1984-1985 Client: CIWTC Location: India Main project features: As part of start up and expansion of river transport operations on the National Waterway No. 1 in the 2nd reach between Patna and Farakka - involved with identification and layout of handling facilities at 8 terminals at Patna, Begusarai, Monghyr, Bhagalpur, Kahalgaon, Sahibganj, Rajmahal and Farakka, . Positions held: Team leader - Port Planner Activities performed: As Project Manager-cum-Planner for development of terminals for the Central Inland Water Corporation (CIWTC), Coordinated site surveys and data collection including traffic surveys, reviewed forecasting, carried out planning and terminal designs and costing, financial projections and tariff assessment. Project Name: Feasibility study for the development of water transport on River Narmada and Narmada Main Canal Year: 1984-1985 Location: India Client: Irrigation and Dam development department, Govt. of Gujarat and International Dutch Assistance program- Den Hague Main project features: The study investigated the feasibility of adding various measures for transfer of ships at 4 dam locations proposed along the River Narmada at Sardar Sarovar, Maheshwar, Omkareshwar and Narmada Sagar. The options considered were - Ship lifts, Hydraulic Inclined planes, locks with options of cascade or a single lift solution and various other water conservation measures. Positions held: Team Member for the solutions at Narmada Sagar dam
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	Activities performed: Involved with both phases of this study concerning river engineering aspects and structural designs for ship transfer and cargo transfer facilities for overcoming water level difference at four major dams along the route with heads up to 100m Project Name: Design, tendering and construction management of 180m x 18m wide dry dock at Rajabagan, Calcutta Year: 1983-1984 Location: India Client: Central Inland Water Corporation (CIWTC), Main project features: Old dry dock with gravity retaining walls was converted into modern dry dock with anchored diaphragm wall construction- 180m x 18m wide with Mitre gate and intermediate log gate. This was part of fleet repair and modernization plan of the operator of river transport services. In addition, construction of a new jetty at the Tea transit sheds for handling of containers moved from Haldia to Kolkata by river barges. Contract envisaged dredging of the dock and its mouth. Positions held: Team Leader-cum-Design Engineer Activities performed: Design, tendering and construction management of dry dock at Rajabagan and a riverside jetty along Hooghly River for Mechanized Handling including containerized cargo carried from Haldia Port. Project Name: Terminal facilities for the development of river terminals in Calcutta, Haldia & Assam Year: 1982-1983 Location: India Client: Central Inland Water Corporation (CIWTC), Main project features: Development of river terminals in Calcutta, Haldia & Assam - Pandu, Jogighopa, Tezpur and Karimganj in Eastern India for the CIWTC Detailed cost estimates, cost flows, IRR etc. were worked out before undertaking detailed engineering and construction of these terminals. Positions held: Team leader - Port Planner Activities performed: Layout and designs of terminal facilities, cargo forecasts, material handling using pallets and containers, interface transfer with the truck terminal etc., Costing. Project Name: Feasibility Study for the development of water transport along Ganga-Bhagirathi-Hooghly River between Allahabad and Farakka, Year: 1981-1982 Location: India Client: Directorate of Transport, Ministry of Shipping and Transport Main project features: Establishing the feasibility of inland water transport on this waterway covering a stretch from Haldia upstream upto Allahabad. - 1650 Km reach. This was one of the major initiatives that led to declaration of Ganga river as the National Waterway no. 1 in 1982. The study envisaged over 20 terminals to be established along the entire route which was to be developed in 3 phases - - Haldia (Sagar)-Farakka)- 560 Km ; - Farakka-Patna- 460 Km - Patna-Allahabad-600 Km Positions held: Project Engineer Activities performed: As a team member assisted different teams involved with terminal designs and river conservancy measures. Name of Project: Development of Gopalpur Port Project in Orissa State, India Year : 1981-1982 Location : India Client: Public Works department, Government of Orissa
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	Main Project Features: Comprising of a 240m long lighterage berth with wall construction, transit sheds, water tanks with roads etc. Position held : Project Engineer Activities Performed: As Project Engineer for the development of Gopalpur Port Project in Orissa State, India Name of Project: Modernization program for Mazagon Docks Ltd Year : 1981-1982 Location : India Client: Mazagon Docks Ltd Main Project Features: Modernization programme Position held : Design Engineer Activities Performed: Responsibilities included design of 8 storey building in RCC housing the offices of the Dock Authority and design of two large span workshops with fabrication facilities for submarine with EOT cranes up to 60T in structural steel. Project Name : SEZ at Mundra port Year : 2008 -09 Client : <i>Adani Port & SEZ Ltd. (public listed co.)</i> Location : Mundra port , Kutchh, Gujarat, India Position Held : President – head of business Main features : One of the largest SEZ in india with an area of over 10,000 Ha to be developed in 2 phases adjacent to Mundra port, Activities performed - Managed a diverse team of over 80 persons covering different functions of marketing, engineering, construction management, regulatory – Legal, corporate liaison, estate, Customer relationship, security, finance. Evolved business plans for the utility company of the group dealing with power distribution, water supply, waste water and sewerage including setting up de-salination plants, water treatment plants and the associated networks. As a Senior management appointee also responsible for the review of costs and development program and its phasing for the Mundra Port and associated infrastructure facilities. Name of assignment or project: Dholera Special Investment Region Year: 2013 Location: India Client: Delhi Mumbai Industrial Corridor development corporation (DMICDC) and the Gujarat state regional development agency. Project Details: The 1st phase development spread over 150 sq. km involving and investment of over 2 Billion USD over 10 years period Positions held: Dy. Program Director Activities performed: Responsible for public consultation process, technical reviews of various projects, formulating engineering and tendering process for this complex project involving partnership between the central government nodal agency – Delhi Mumbai Industrial Corridor development corporation (DMICDC) and the Gujarat state regional development agency. Name of assignment or project : <i>Brahma City project</i> Year :2012-13 Client : Brahma Reality Pvt. Ltd. Location : sector 61, 63, 65 Gurgaon, India Position Held : <i>Project Director</i> Main features : A 180 acre development spread over 4 land parcels in different
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	sectors of Gurgaon. Integrated mixed use development with low rise township, high rise block and commercial centre Activities performed : planning and design of the infrastructure services including roads, water and waste water, STP, electrical distribution network and transformer station, solid waste management Project Name : Master Engineering of DLF - Phase 5 Year : 2012-13 Client : DLF Ltd. Location : Gurgaon, India Position Held : <i>Project Director</i> Main features : Centre around the Golf course in Gurgaon it covers planning and engineering activities over an area of about 400 acres., re-generation and other green city development concepts. Activities performed : Created a bench-mark / reference of best practices related to planning and engineering of the infrastructure like roads, water supply, drainage, wastewater networks, treatment, re-cycling, electrical distribution and back up generation, energy conservation, demand management. Project Name : Urban renewal project of DLF 1 – 4, Gurgaon Year : 2012 Client : DLF Ltd. Location : Gurgaon, india Position Held : Project Director Main features : Urban renewal project covering the 4 sectors of leading developer of India – DLF covering over 900 acres of land. Based on stake holder surveys and participation. Activities performed : . review of streetscape, traffic forecasting, flow analysis of transport networks, junction improvements, Landscaping, signage and undertaking holistic analysis of water supply, sewerage, drainage networks with the aim of incorporating re-cycling, ground water re-charge, re-generation of water bodies. On the power supply and distribution – evaluation of demand management measures with community Name of assignment or project: 1400 acre development for Omega City project Year: 2012-13 Location: India Client: Apricot Realtors Private Limited Project Details: industrial, institutional and Golf based residential development Positions held: Project Director Activities performed: Responsible for planning and design of infrastructure facilities inclusive of roads, bridges, water, waste water, storm drainage, CETP, STP, Electrical distribution and solid waste management system. Name of Project: Township for NTPC Koldam Project Year : 2002-2003 Location : India Client: NTPC with DKS Alplan Pvt. Ltd Main Project Features : The Architecture, Planning and Design Position held : Team leader
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	Activities Performed: Responsible for the coordination of the work and Client deliverables for a Township project spread over a land area of 60 acres near Bilaspur in Himachal Pradesh.

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Certification:

I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience, and I am available to undertake the assignment in case of an award. I understand that any misstatement or misrepresentation described herein may lead to my disqualification or dismissal by the Client, and/or sanctions by the Bank.

Rajinder Khazanchi



1 Nov 2022

Name of Expert

Signature

Date